



МИНОБРНАУКИ РОССИИ
Федеральное государственное бюджетное образовательное учреждение
высшего образования
«МИРЭА – Российский технологический университет»
РТУ МИРЭА

Контрольная работа № 1

Выполнил студент 2 курса группы
ЭФБО-04-24 Валекжанин В.С.

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Примечания

Данные примечания продублированы из README.md для удобства.

В скриншотах в директории ``report/`` выполняются стандартные команды, но с помощью псевдонимов. Вот все псевдонимы и их значение:

- ``mcr`` - ``micro`` (CLI-редактор файлов)
- ``gm`` - ``git merge``
- ``gchk`` - ``git checkout``
- ``gcl`` - ``git clone``
- ``ga`` - ``git add``
- ``gcmmsg <message>`` - ``git commit -m <message>``
- ``push`` - ``git push``

Скриншоты

```

valdemar_check:spo-kr1/ (main) $ ls [11:13:36]
.      ..      .git      LICENSE
valdemar_check:spo-kr1/ (main) $ gchk -b test_branch [11:13:36]
Switched to a new branch 'test_branch'
valdemar_check:spo-kr1/ (test_branch) $ mkdir KR1 [11:13:52]
valdemar_check:spo-kr1/ (test_branch) $ cd KR1 [11:14:16]
valdemar_check:KR1/ (test_branch) $ touch hello.h hello.cpp main.cpp Makefile
valdemar_check:KR1/ (test_branchx) $ alias mcr=micro [11:16:07]
valdemar_check:KR1/ (test_branchx) $ mcr hello.h [11:16:14]
valdemar_check:KR1/ (test_branchx) $ mcr hello.cpp [11:16:35]
valdemar_check:KR1/ (test_branchx) $ mcr main.cpp [11:16:45]
valdemar_check:KR1/ (test_branchx) $ mcr Makefile [11:16:57]
valdemar_check:KR1/ (test_branchx) $ make [11:17:05]

valdemar_check:KR1/ (test_branchx) $ ls [11:17:28]
.      ..      Makefile  hello.cpp  hello.h    main.cpp
valdemar_check:KR1/ (test_branchx) $ ls -l [11:17:29]
Found existing alias for "ls -l". You should use: "ll"
total 32
drwxr-xr-x  6 valdemar_check  staff  192 Feb 26 11:14 .
drwxr-xr-x  5 valdemar_check  staff  160 Feb 26 11:14 ..
-rw-r--r--  1 valdemar_check  staff  254 Feb 26 11:17 Makefile
-rw-r--r--  1 valdemar_check  staff   94 Feb 26 11:16 hello.cpp
-rw-r--r--  1 valdemar_check  staff  102 Feb 26 11:16 hello.h
-rw-r--r--  1 valdemar_check  staff  101 Feb 26 11:16 main.cpp
valdemar_check:KR1/ (test_branchx) $ make [11:17:31]
g++ -std=c++11 -Wall -Wextra -c main.cpp -o main.o
g++ -std=c++11 -Wall -Wextra -c hello.cpp -o hello.o
g++ -std=c++11 -Wall -Wextra -o app main.o hello.o
valdemar_check:KR1/ (test_branchx) $ ls [11:17:34]
.      Makefile  hello.cpp  hello.o    main.o
..      app      hello.h    main.cpp
valdemar_check:KR1/ (test_branchx) $ ./app [11:17:36]
Hello, World!
valdemar_check:KR1/ (test_branchx) $ make clean [11:17:38]
rm -f main.o hello.o app
valdemar_check:KR1/ (test_branchx) $ ls [11:17:42]
.      ..      Makefile  hello.cpp  hello.h    main.cpp

```

```

valdemar_check:SPO/ $ cd spo-kr1 [11:08:29] |
valdemar_check:spo-kr1/ $ gh repo create [11:08:32] |
? What would you like to do? Create a new repository on github.com from s
cratch
[?] Repository name spo-kr1 |
[?] Repository owner vladcheck |
[?] Description |
[?] Visibility Public |
[?] Would you like to add a README file? No |
[?] Would you like to add a .gitignore? No |
[?] Would you like to add a license? Yes |
[?] Choose a license MIT License |
[?] This will create "spo-kr1" as a public repository on github.com. Contin
ue? Yes
✓ Created repository vladcheck/spo-kr1 on github.com
https://github.com/vladcheck/spo-kr1
[?] Clone the new repository locally? No |

```

```

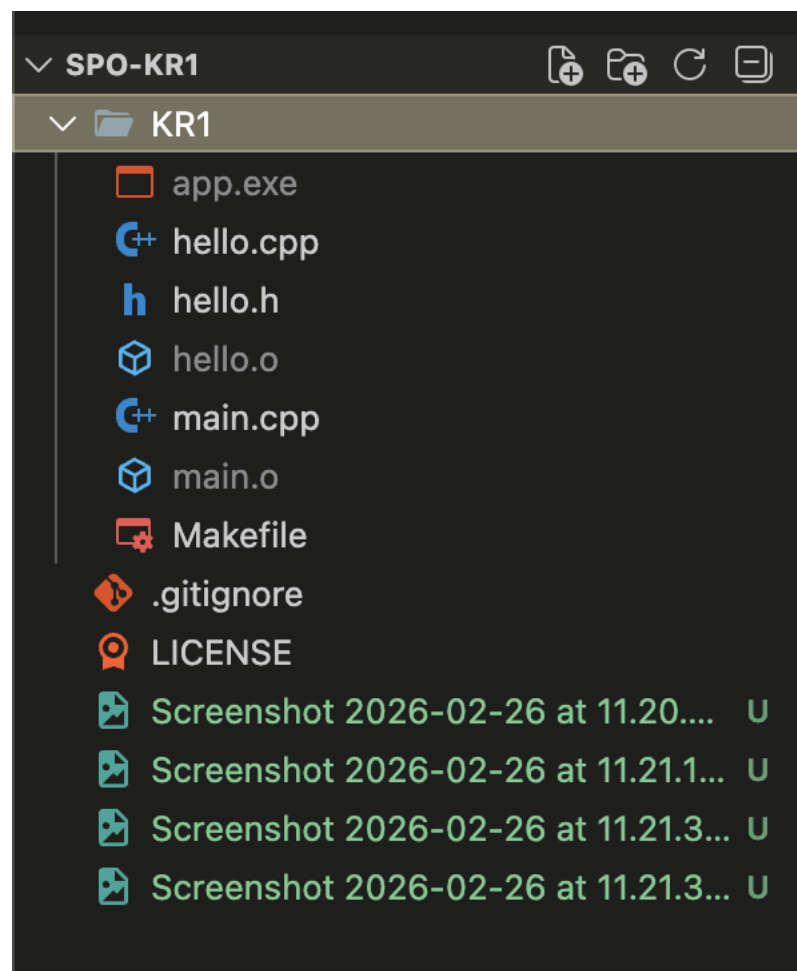
valdemar_check:KR1/ (test_branchx) $ mcr Makefile [11:17:44]
valdemar_check:KR1/ (test_branchx) $ make [11:17:59]
g++ -std=c++11 -Wall -Wextra -c main.cpp -o main.o
g++ -std=c++11 -Wall -Wextra -c hello.cpp -o hello.o
g++ -std=c++11 -Wall -Wextra -o app.exe main.o hello.o
valdemar_check:KR1/ (test_branchx) $ ls [11:18:02]
.      Makefile  hello.cpp  hello.o   main.o
..     app.exe   hello.h   main.cpp
valdemar_check:KR1/ (test_branchx) $ make clean [11:18:03]
rm -f main.o hello.o app.exe
valdemar_check:KR1/ (test_branchx) $ .. [11:18:07]
valdemar_check:spo-kr1/ (test_branchx) $ micro .gitignore [11:18:09]
Found existing alias for "micro". You should use: "mcr"
valdemar_check:spo-kr1/ (test_branchx) $ cd KR1 [11:18:18]
valdemar_check:KR1/ (test_branchx) $ make [11:18:21]
g++ -std=c++11 -Wall -Wextra -c main.cpp -o main.o
g++ -std=c++11 -Wall -Wextra -c hello.cpp -o hello.o
g++ -std=c++11 -Wall -Wextra -o app.exe main.o hello.o
valdemar_check:KR1/ (test_branchx) $ ls [11:18:24]
.      Makefile  hello.cpp  hello.o   main.o
..     app.exe   hello.h   main.cpp
valdemar_check:KR1/ (test_branchx) $ .. [11:18:24]
valdemar_check:spo-kr1/ (test_branchx) $ ga . [11:18:27]
valdemar_check:spo-kr1/ (test_branchx) $ gcmmsg "feat: create helloWorld program"
[test_branch 2057c2d] feat: create helloWorld program
5 files changed, 39 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 KR1/Makefile
 create mode 100644 KR1/hello.cpp
 create mode 100644 KR1/hello.h
 create mode 100644 KR1/main.cpp
valdemar_check:spo-kr1/ (test_branch) $ push [11:18:35]
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 10 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 932 bytes | 932.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'test_branch' on GitHub by visiting:
remote: https://github.com/vladcheck/spo-kr1/pull/new/test_branch

```

```

create mode 100644 KR1/main.cpp
valdemar_check:spo-kr1/ (test_branch) $ push [11:18:35]
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 10 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 932 bytes | 932.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'test_branch' on GitHub by visiting:
remote: https://github.com/vladcheck/spo-kr1/pull/new/test_branch
remote:
To github.com:vladcheck/spo-kr1.git
 * [new branch] test_branch -> test_branch
branch 'test_branch' set up to track 'origin/test_branch'.
valdemar_check:spo-kr1/ (test_branch) $ gchk main [11:18:52]
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
valdemar_check:spo-kr1/ (main) $ type gm [11:18:58]
gm is an alias for git merge
valdemar_check:spo-kr1/ (main) $ gm test_branch [11:19:05]
Updating 33f6d6d..2057c2d
Fast-forward
 .gitignore      | 2 ++
 KR1/Makefile    | 17 ++++++
 KR1/hello.cpp   | 5 +++++
 KR1/hello.h     | 8 ++++++
 KR1/main.cpp    | 7 ++++++
 5 files changed, 39 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 KR1/Makefile
 create mode 100644 KR1/hello.cpp
 create mode 100644 KR1/hello.h
 create mode 100644 KR1/main.cpp
valdemar_check:spo-kr1/ (main) $ push [11:19:11]
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:vladcheck/spo-kr1.git
 33f6d6d..2057c2d main -> main
valdemar_check:spo-kr1/ (main) $

```



```
hello.cpp x
KR1 > hello.cpp
1  #include "hello.h"
2
3  void printHelloWorld(std::ostream& out) {
4      out << "Hello, World!\n";
5  }
6
```

```
main.cpp x
KR1 > main.cpp
1  #include <iostream>
2  #include "hello.h"
3
4  int main() {
5      printHelloWorld(std::cout);
6      return 0;
7  }
8
```


h hello.h ×

KR1 > h hello.h

```
1  #ifndef HELLO_H
2  #define HELLO_H
3
4  #include <ostream>
5
6  void printHelloWorld(std::ostream& out);
7
8  #endif
9
```

Makefile

KR1 > Makefile

```
1 CXX = g++
2 CXXFLAGS = -std=c++11 -Wall -Wextra
3 TARGET = app.exe
4 OBJS = main.o hello.o
5
6 all: $(TARGET)
7
8 $(TARGET): $(OBJS)
9     $(CXX) $(CXXFLAGS) -o $(TARGET) $(OBJS)
10
11 %.o: %.cpp
12     $(CXX) $(CXXFLAGS) -c $< -o $@
13
14 clean:
15     rm -f $(OBJS) $(TARGET)
16
17 .PHONY: all clean
18
```